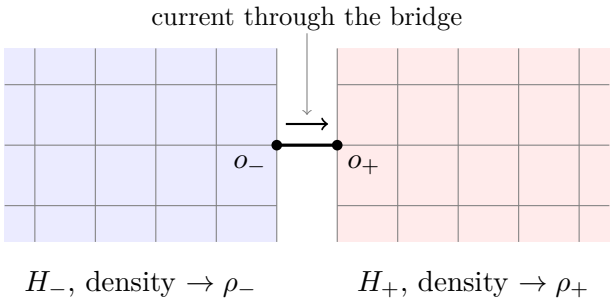
Symmetric exclusion on two half-spaces of $\mathbb{Z}^3$: a non-reversible stationary measure

counterexamples-ips, entry 0002

Status: *plausible* (not yet checked by a human)
Counterexample: ChatGPT Astra-6 (Section 8 of the manuscript in `raw/`)
Digest: Claude (Opus 5.5), 2 November 2026

Question. For exchange dynamics on a connected graph of bounded degree, with positive rates on every edge: is every stationary measure with finite energy reversible? On recurrent graphs (in particular $\mathbb{Z}^d$, $d \le 2$) the answer is *yes*, by the conservative analogue of Holley–Stroock (Section 7 of the source). On transient graphs the answer is *no*.

Setting. Let $H_+ = \{x \in \mathbb{Z}^3 : x_1 \ge 1\}$, $H_- = \{x \in \mathbb{Z}^3 : x_1 \le 0\}$, $o_- = (0,0,0)$, $o_+ = (1,0,0)$. The graph G on $\mathbb{Z}^3$ has as edges all nearest-neighbour bonds inside H_+ or inside H_-, and the single bridge $\{o_-, o_+\}$.



SEP on G exchanges the spins at the endpoints of each edge of G at rate 1. A measure is reversible iff it is invariant under each exchange θ_b, b an edge of G; all Bernoulli product measures ν_ρ are reversible.

Theorem 1 (Theorem 8.1 of the source)**.** *Let $0 < \rho_- < \rho_+ < 1$. SEP on G has a stationary measure μ^* that is not reversible and has finite energy with constant $\delta = \min\{\rho_-, 1-\rho_+\}$. Moreover $\mu^*(\eta_x = +1) \to \rho_\pm$ as $|x| \to \infty$ in $H_\pm$.*

Proof (summary). Step 1. Simple random walk X on G is transient (fold $\mathbb{Z}^3$ along $\{x_1 = \frac{1}{2}\}$; Rayleigh monotonicity), so it crosses the bridge finitely often, and

$$\alpha(x) := P_x(X \text{ eventually stays in } H_+)$$

is harmonic with $0 \le \alpha \le 1$. By the reflection symmetry $\alpha(1-x_1, x_2, x_3) = 1 - \alpha(x)$, and $\alpha > \frac{1}{2}$ far out in H_+; so α is not constant. Put $\beta = \rho_- + (\rho_+ - \rho_-)\alpha$.

Step 2. Start from the product measure ν_β. The one-point function $u_t(x)$ of SEP solves $\partial_t u = \Delta_G u$; since β is harmonic, $u_t = \beta$ for all t.

Step 3. A weak limit μ^* of Cesàro averages of $\nu_\beta T_t$ is stationary, with $\mu^*(\eta_x = +1) = \beta(x)$.

Step 4. Reversibility would force $\beta(x) = \beta(y)$ along every edge, so β would be constant on the connected graph G.

Step 5. By the stirring construction $\eta_t = \eta_0 \circ \pi_t^{-1}$ with π_t independent of η_0; conditionally on π_t the spins are independent with densities in $[\rho_-, \rho_+]$, which gives finite energy. □

Mechanism. The half-spaces act as two reservoirs at densities ρ_- and ρ_+ joined by one wire. Because they are transient, particles that cross the bridge wander off rather than equilibrating the two sides, so the density difference persists and a current flows through the bridge in the stationary state. A current is incompatible with reversibility. On recurrent graphs bounded harmonic functions are constant and every stationary measure of SEP is a mixture of Bernoulli measures (Liggett's characterisation).

Remark (Interacting version, Theorem 8.3 of the source; not checked). For exchange rates on G associated with a potential J with $\sum_F |J_F| < \infty$, there is a reversible measure ($\propto e^{-\Phi}\nu_\rho$) and also a stationary measure μ with $|\mu(\eta_x = +1) - \beta(x)| \le \gamma(x) \to 0$, hence not reversible. Proof by coupling with SEP. Whether μ has finite energy is open.

Which hypothesis breaks. (i) G is not $\mathbb{Z}^3$: rates vanish on all bonds crossing $\{x_1 = \frac{1}{2}\}$ except one. With rates bounded away from 0 on all bonds of $\mathbb{Z}^3$, bounded harmonic functions are constant (elliptic Harnack) and all stationary measures of SEP are reversible. (ii) The dynamics is conservative; for spin flips no analogue is known, so the Holley–Stroock question in $d \ge 3$ remains open. (iii) The example shows that recurrence cannot be dropped from the graph version of the conservative Holley–Stroock theorem.

Follow-up questions. See `entries/0002-sep-two-half-spaces-z3/followups.md`.

References

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